Sahil

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Summary

Computer Science undergraduate with strong foundations in **DSA, C++, Python, and SQL**, and hands-on experience building backend applications and data pipelines.

Education

Vellore Institute of Technology Sep 2023 – 2027

*B.Tech in Computer Science and Engineering – CGPA: **9.05***

Technical Skills

Languages: C++, Python

Frontend: React, TypeScript, HTML, CSS

Backend & APIs: FastAPI, Flask, REST APIs

Data & Databases: SQL, PostgreSQL, Pandas, Data Modeling

Tools & Infrastructure: Git, Docker, Docker Compose, Nginx, AWS

Projects

MANDIBHAV: Agricultural Price Analytics Platform GitHub

Python, SQL, PostgreSQL, Pandas, Metabase

- Built a **Python** pipeline to fetch daily government data from a **REST API** and store it in a **PostgreSQL star schema** with Fact and Dimension tables, adding retry logic for failed requests.
- Used **Pandas** in **Jupyter Notebooks** for **Exploratory Data Analysis (EDA)** to inspect price ranges, handle missing values, and filter out data entry errors before saving to the database.
- Wrote **SQL Window Functions (LAG, RANK, STDDEV)** to calculate 30-day price volatility and compare market prices against government **MSP** baselines.
- Built an interactive **Metabase** dashboard, deployed using **Docker**, to display price trends, compare markets across states, and flag when crops sell below MSP.

Resume Builder: Containerized Web Application GitHub | Live Demo

React, TypeScript, Python, Flask, Docker, Gemini API

- Built a full-stack web app using **React/TypeScript** and **Flask**, running services in separate **Docker** containers and using **Nginx** to route API requests.
- Integrated the **Gemini API** to review resumes, extract required skills from job descriptions for match scoring, and provide an in-editor AI assistant for real-time tips.
- Built a secure LaTeX compiler workflow inside **Docker** with a **10-second timeout** to stop infinite loops and automatic cleanup to prevent leftover build files.
- Created a dual-pane editor using **Monaco Editor** with a **1.5-second debounce** to cut unnecessary compilation calls by **˜80%** and proper memory cleanup for generated PDFs.

Achievements

- Solved **600+ problems on** LeetCode and **100+ problems on** Codeforces.
- Ranked among the **Top 100 teams nationwide** in the **ZS Campus Beats Tech Challenge (2026)**.
- Ranked among the **Top 10 teams in the state** at the AMD Slingshot Hackathon; built **Zenith**, a developer security tool to detect hardcoded secrets and vulnerable dependencies (VS Code Extension).

Open Source Contribution

Microsoft SQL Tools Service Merged PR #2804

- Fixed an Object Explorer bug that labeled Columnstore Indexes as ordinary clustered indexes by updating the index-type check and label text.
- Added unit tests for the new Columnstore labels and regression tests for existing labels.

Certifications

AWS Certified Solutions Architect – Associate